

Problem

Given that

$$[g] = \begin{bmatrix} 0.06 \text{ S} & -0.4 \\ 0.2 & 2 \Omega \end{bmatrix}$$

determine:

(a) $[z]$

(b) $[y]$

(c) $[h]$

(d) $[T]$

Step-by-step solution

Step 1 of 5

We have,

$$[g] = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} 0.06 \text{ S} & -0.4 \\ 0.2 & 2 \Omega \end{bmatrix}$$

$$\text{so, } \Delta g = g_{11}g_{22} - g_{21}g_{12} = 0.2$$

Step 2 of 5

$$(a) [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{g_{11}} & \frac{-g_{12}}{g_{11}} \\ \frac{g_{21}}{g_{11}} & \frac{\Delta g}{g_{11}} \end{bmatrix} = \begin{bmatrix} \left(\frac{1}{0.06 \text{ S}} \right) & -\left(\frac{-0.4}{0.06 \text{ S}} \right) \\ \left(\frac{0.2}{0.06 \text{ S}} \right) & \left(\frac{0.2}{0.06 \text{ S}} \right) \end{bmatrix}$$

$$[Z] = \begin{bmatrix} 16.667 \Omega & 6.667 \Omega \\ 3.333 \Omega & 3.333 \Omega \end{bmatrix}$$

Step 3 of 5

$$(b) [y] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \frac{1}{g_{22}} \begin{bmatrix} \Delta g & g_{12} \\ -g_{21} & 1 \end{bmatrix} = \frac{1}{(2 \Omega)} \begin{bmatrix} 0.2 & -0.4 \\ -0.20 & 1 \end{bmatrix}$$

$$[y] = \begin{bmatrix} 0.1 & -0.2 \\ -0.1 & 0.5 \end{bmatrix} \text{ S}$$

Step 4 of 5

$$(c) [h] = \frac{1}{\Delta g} \begin{bmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{bmatrix} = \frac{1}{(0.2)} \begin{bmatrix} 2 & -(-0.4) \\ -0.2 & 0.06 \end{bmatrix} = \begin{bmatrix} 10 \Omega & 2 \\ -1 & 0.3 \text{ S} \end{bmatrix}$$

Step 5 of 5

$$(d) \quad [T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \frac{1}{g_{11}} \begin{bmatrix} 1 & g_{22} \\ g_{11} & \Delta g \end{bmatrix} = \frac{1}{(0.2)} \begin{bmatrix} 1 & 2 \\ 0.06 & 0.2 \end{bmatrix} = \begin{bmatrix} 5 & 10\Omega \\ 0.3s & 1 \end{bmatrix}$$